

M004 · Your Function Library — Reusable Building Blocks — English Worksheet

Topic: Your Function Library — Reusable Building Blocks · **Course:** Functions & Games · **Time:** about 45 minutes

This lesson you collect your best functions into a **library** — reusable building blocks like `build_wall`, `build_tower`, and `clear_area`. Big builds become easy when small functions do the work.

1 · Predict

```
function build_wall(length):
    repeat length times:
        place stone block
        move forward 1

build_wall(5)
build_wall(3)
```

The function is called twice with different numbers. How many stone blocks get placed in total? Which call builds the longer wall?


```
function build_tower(height):
    repeat height times:
        place stone block
        move up 1

function build_house():
    build_wall(4)
    build_tower(3)

build_house()
```

build_house calls two helper functions. What gets built, and in what order?


```
function clear_area(size):  
    repeat size times:  
        repeat size times:  
            replace block with air  
            move forward 1  
        move right 1
```

```
clear_area(4)  
build_wall(4)
```

What happens first — clearing or building? Why does the order of these two calls matter?

2 · Spot the Bug 🐛

Each block below is broken. Rewrite it so it works, then explain why the original was wrong and why your fix works.

Bug A — The library should have **two different** functions: a stone wall and a glass wall. Right now both functions have the **same name**, so the computer gets confused about which one to use.

```
function build_wall(length):  
  repeat length times:  
    place stone block  
    move forward 1  
  
function build_wall(length):  
  repeat length times:  
    place glass block  
    move forward 1  
  
build_wall(4)
```

Hint: every function in your library needs its own clear name.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — `build_house` is supposed to build four walls **and a door**. The door function exists, but the house never gets a door.

```

function build_door():
    replace block with air
    replace block above with air

function build_house():
    build_wall(4)
    build_wall(4)
    build_wall(4)
    build_wall(4)

build_house()

```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — `build_tower(height)` should build a tower **as tall as the number you give it**. But `build_tower(10)` keeps making a tiny tower.

```

function build_tower(height):
    repeat 3 times:
        place stone block
        move up 1

build_tower(10)

```

Hint: the parameter `height` is never used inside the function.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build something using your own function library — one function should call another function.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.